

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE STRUCTURE

Under New Education Policy (NEP) 2020: Academic Year 2026-2027 and Onwards

B.Tech. (Computer Science and Engineering) Specialization in Artificial Intelligence



After Year 1: U.G. After Year 2: U.G. Diploma After Year 3: B.Sc. (Computer Science) After Year 4: B.Tech. Degree
Certificate

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SEMESTER I									
S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping
1	CA101	Problem Solving Using C	3	0	0	3	PCC	Professional Core Course	Major (DSC)
2	ECE101	Electronics and Electrical Engineering	3	0	0	3	PCC	Engineering Science Course	Major (DSC)
3	MA102	Engineering Physics I	3	1	0	4	PCC	Basic Science Course	Major (DSC)
4	MA111	Applied Mathematics I	3	1	0	4	PCC	Basic Science Course	Major (DSC)
5	ICT101	Career Skills	2	0	0	2	PCC	Basic Science Course	Major (DSC)
6	PH104	Engineering Physics Lab	0	0	2	1	PCC	Basic Science Course	Major (DSC)
7	ICT181	ICT Workshop I	0	0	2	1	PCC	Engineering Science Course	Major (DSC)
8	CA181	Problem Solving Using C Lab	0	0	2	1	PCC	Professional Core Course	Major (DSC)
9	ECE185	Electronics and Electrical Engineering Lab	0	0	2	1	PCC	Engineering Science Course	Major (DSC)
10	XXXX	Slot for HSMC/VAC	2	0	0	2	HSMC	Humanities Social Science and Management Course	Value Added Course (VAC)
Total Hours and Credits			16	2	8	22			

SEMESTER II									
S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping
1	CA102	Python Programming	3	0	0	3	PCC	Professional Core Course	Major (DSC)
2	MA112	Applied Mathematics II	3	1	0	4	PCC	Basic Science Course	Major (DSC)
3	CA104	Introduction to Artificial Intelligence	2	0	0	2	PCC	Professional Core Course	Major (DSC)
4	ICT102	Discrete Mathematics	3	0	0	3	PCC	Basic Science Course	Major (DSC)
5	ES101	Environmental Studies	3	1	0	4	PCC	Basic Science Course	Major (DSC)
6	EN101	English Proficiency	2	0	0	2	HSMC	Humanities Social Science and Management Course	Major (DSC)
7	CA182	Python Programming Lab	0	0	2	1	PCC	Professional Core Course	Major (DSC)
8	ICT182	ICT Workshop II	0	0	2	1	PCC	Engineering Science Course	Major (DSC)
9	ICT184	Career Skills Lab	0	0	2	1	HSMC	Humanities Social Science and Management Course	Major (DSC)
10	XXXX	Slot for HSMC/VAC	2	0	0	2	HSMC	Humanities Social Science and Management Course	Value Added Course (VAC)
Total Hours and Credits			18	2	6	23			

SEMESTER III									
S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping
1	CA103	Scientific Computing with Python	3	0	0	3	PCC	Professional Core Course	Major (DSC)
2	CA103	Data Structures	3	0	0	3	PCC	Professional Core Course	Major (DSC)
3	CA105	Machine Learning	3	0	0	3	PCC	Professional Core Course	Major (DSC)
4	CA107	Enterprise Networks	3	0	0	3	PCC	Professional Core Course	Major (DSC)
5	CA106	Operating Systems	3	0	0	3	PCC	Professional Core Course	Major (DSC)
6	CA108	Scientific Computing with Python Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
7	CA183	Data Structures Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
8	CA183	Computer Networks Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
9	CA187	Machine Learning Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
Total Hours and Credits			15	0	16	23			

SEMESTER IV									
S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping	UCC Mapping	UCC Mapping
1	CA109	Design and Analysis of Algorithms	3	0	0	3	PCC	Professional Core Course	Major (DSC)
2	CA110	Database Systems	3	0	0	3	PCC	Professional Core Course	Major (DSC)
3	CA111	Security in Networks	3	0	0	3	PCC	Professional Core Course	Major (DSC)
4	CA112	Cloud Computing	3	0	0	3	PCC	Professional Core Course	Major (DSC)
5	CA113	Big Data Analytics	3	0	0	3	PCC	Professional Core Course	Major (DSC)
6	CA114	Design and Analysis of Algorithms Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
7	CA115	Database Systems Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
8	CA116	Security in Networks Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
9	CA117	Cloud Computing Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
10	CA118	Big Data Analytics Lab	0	0	4	2	PCC	Professional Core Course	Major (DSC)
Total Hours and Credits			15	0	16	23			






